



SPATIAL ANALYSIS LAB
UNIVERSITY OF RICHMOND

Mapping Heat and Planting Shade: Cooling Historic Jackson Ward

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Introduction and Overview

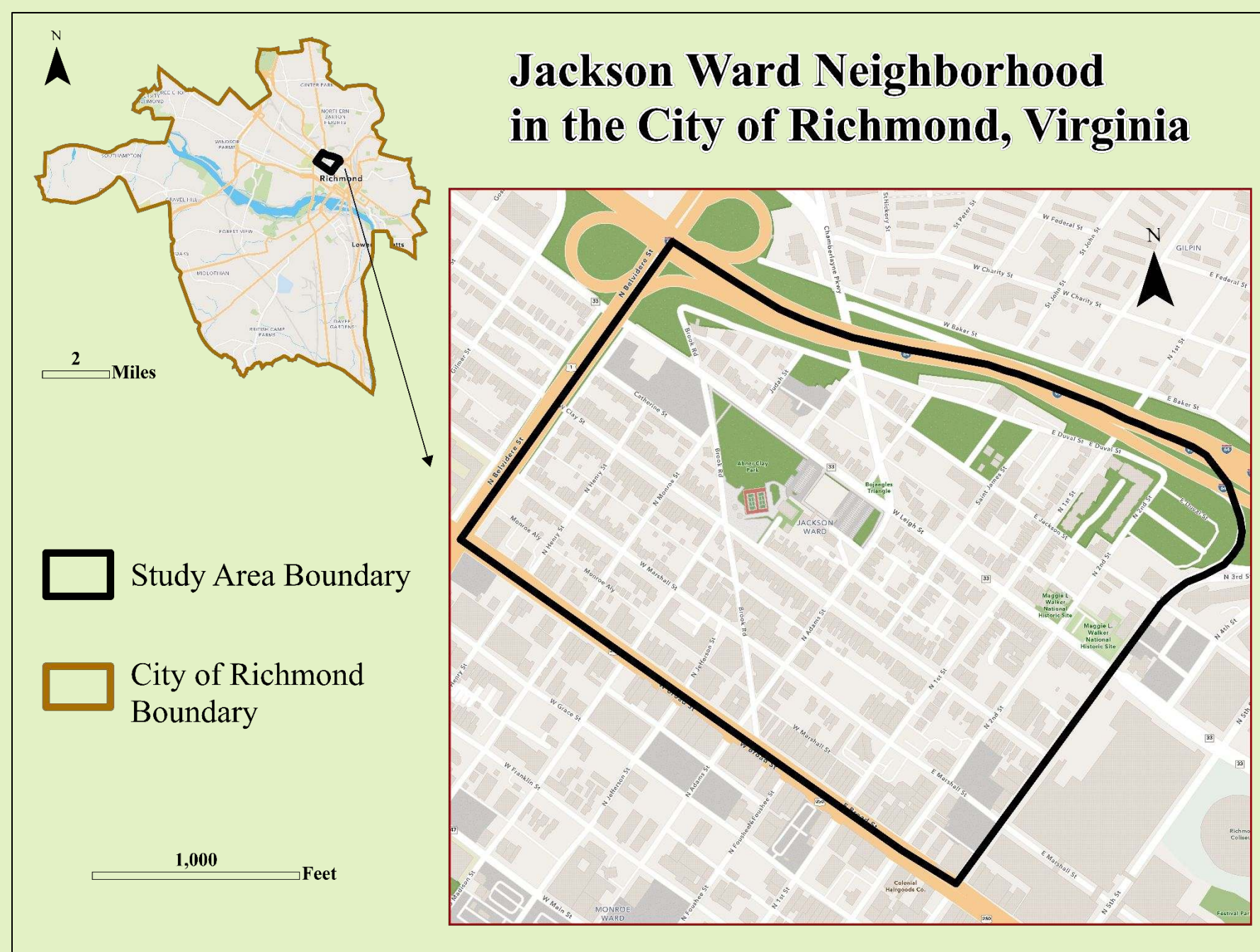


Figure 1: Map of the Jackson Ward study area in downtown Richmond, Virginia

- Excessive heat has become a significant public health issue in Jackson Ward, a historically Black neighborhood near downtown Richmond, Virginia (Braun *et al.*, 2024).
 - From the 1930s, the neighborhood's urban landscape has been shaped by discriminatory redlining, divisive I-95 highway construction and extensive urban renewal (Hoffman *et al.*, 2020).
- Intra-neighborhood heterogeneity in sidewalk heat reveals distinct microclimates across the neighborhood and unevenness in heat risk.
 - This work enables spatially targeted mitigation strategies to combat pedestrian heat risk.

Study Objectives

1. Collaborate with the Historic Jackson Ward Association to address the neighborhood's increasingly disproportionate heat-related health burden
2. Strategically plant trees in vacant wells around Jackson Ward to enhance shade and reduce pedestrian heat stress
3. Encourage sustainable, long-term greening activities that promote climate resilience throughout Jackson Ward
4. Monitor and document the potential benefits of tree planting on pedestrian microclimate variability and community wellbeing in urban environments

Heat Data Collection Process

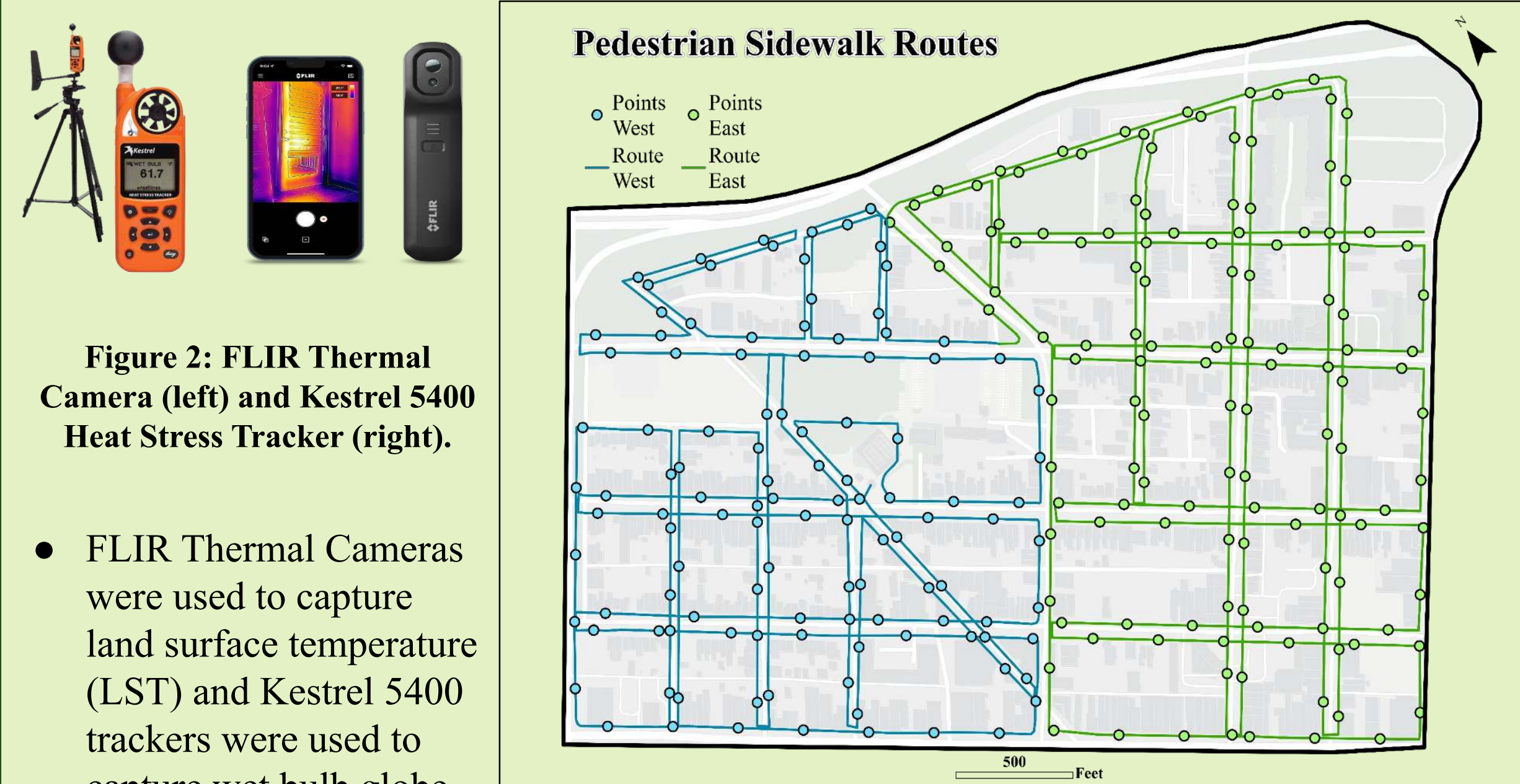


Figure 3: Map of pedestrian sidewalk routes and data collection stops

- FLIR Thermal Cameras were used to capture land surface temperature (LST) and Kestrel 5400 trackers were used to capture wet bulb globe temperature (WBGT).
- Study conducted during peak 2025 summer heat hours from 12pm-4pm across five days - June 10th, 11th, 20th, 21st and 24th.
- Temperature was recorded at points along 85-meter intervals, then interpolated to a continuous surface using Inverse Distance Weighting (IDW) and standardized to account for diurnal variation in ambient temperature during data collection.

Heat Threat and Planting Sites

Heat Threat Level for Pedestrians in Jackson Ward on June 21st 2025

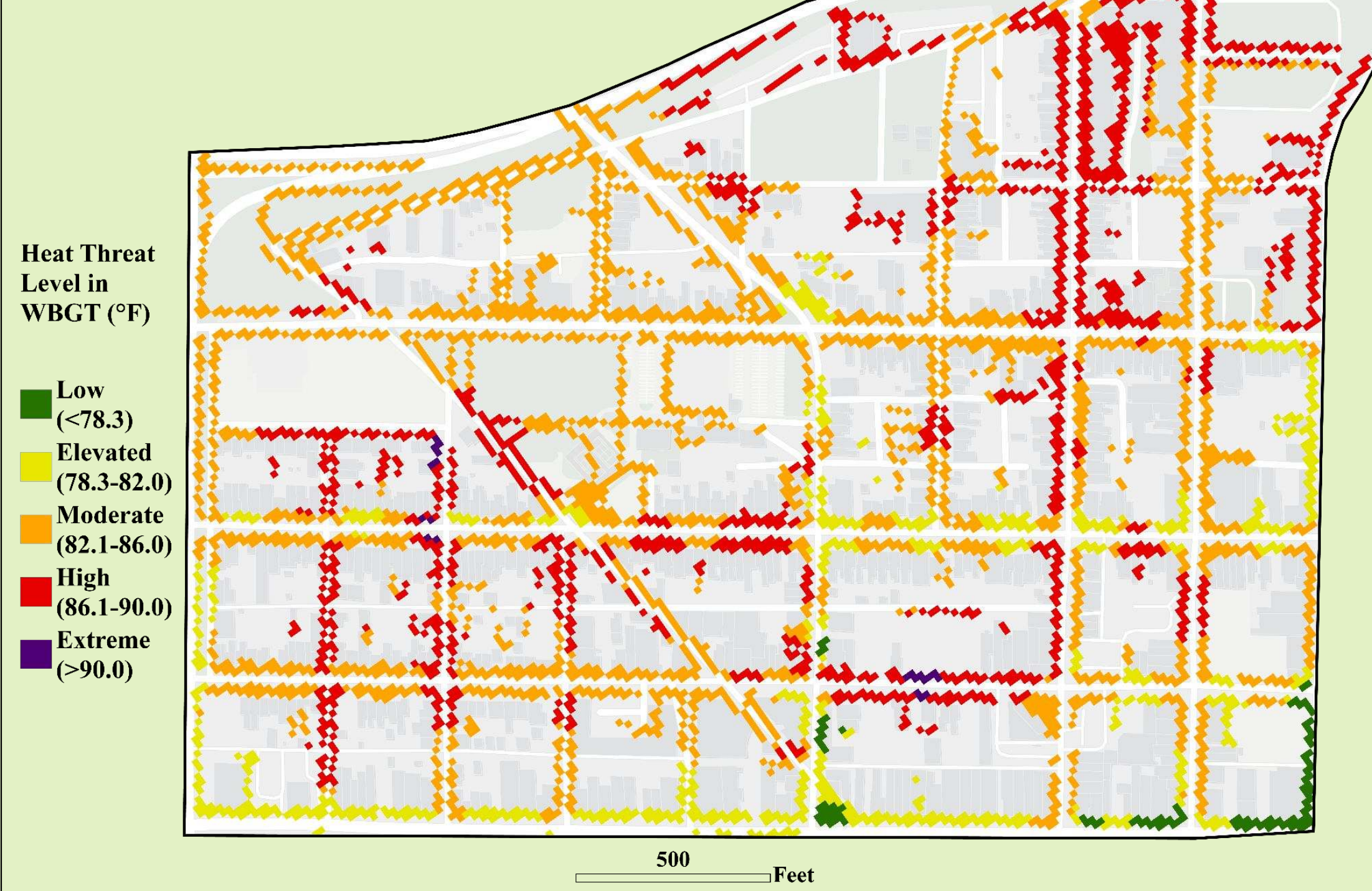
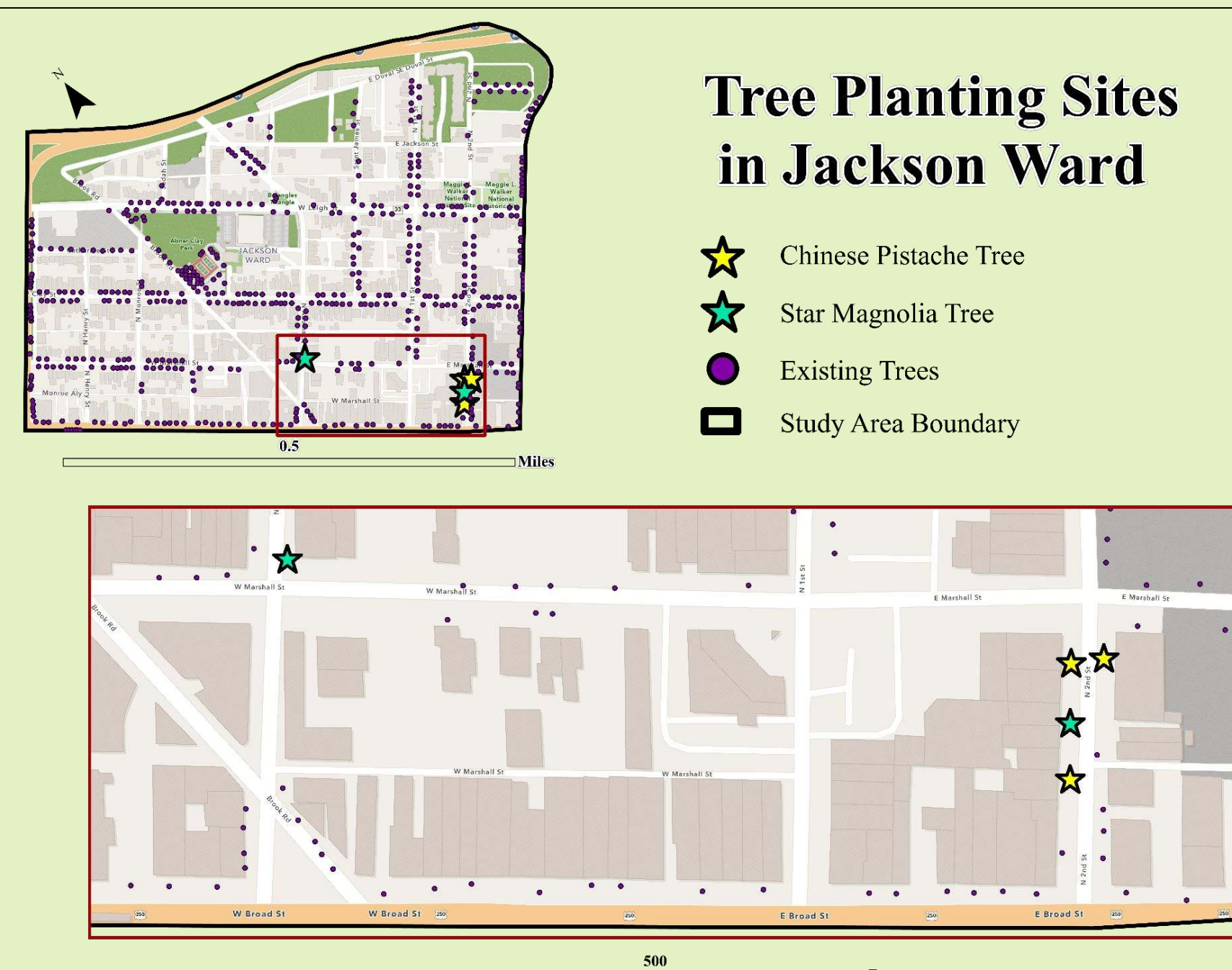


Figure 4: Map of heat threat levels across sidewalks of Jackson Ward in Richmond, Virginia.

- Aggregating the heat data allowed us to identify relative hotspots across the neighborhood as priority areas for tree planting.
- The hottest area of Jackson Ward is concentrated on West Marshall Street, located between 1st and 2nd Streets on the industrialized east section with noted minimal tree canopy (see Figure 10).
- Hotspot areas were then overlaid with the City of Richmond's existing tree well inventory to identify planting opportunities within or near the highest heat burden locations.
- Five tree wells were identified for planting, including three Chinese pistache (*Pistacia chinensis*) and two star magnolias (*Magnolia stellata*).

Figure 5: Map of priority tree wells identified in Jackson Ward

Tree Planting Sites in Jackson Ward



Tree Species

- Tree species were recommended by Michael Webb, Urban Forester at the City of Richmond Department of Public Works, based on the size of existing tree wells and performance in urban environments.
- Chinese pistache and star magnolia were selected for their urban durability and low long-term maintenance requirements.
- Species selection also incorporated the Jackson Ward community's wants for the types of trees.



Figure 6 (left):
Image of a star
magnolia tree
(*Magnolia stellata*)



Figure 7
(right): Image
of a Chinese
pistache tree
(*Pistacia chinensis*)

Cultural Homage



Figure 8: Image of a few plaques



Figure 9: Image of erected tree with Maggie Walker plaque

- Each of the five planted trees is accompanied by a plaque featuring a quote from a historical icon with roots in Jackson Ward.
- These plaques honor the figures whose legacy earned the neighborhood its reputation as the "Harlem of the South".
- Quotes draw from icons including Mrs. Maggie L. Walker and Rev. William Washington Browne, grounding the environmental intervention in the neighborhood's rich cultural heritage.

Contrasting Thermal Streetscapes



Figure 10: Hottest Identified Sidewalk (A) and Coolest Identified Sidewalk (B) in Jackson Ward. The hottest spot is located on W Marshall St. between 1st St. and N Adams St. The coolest spot is also located on W Marshall St., between N Monroe St. and N. Madison St.

- On average, the hottest spot (A) had a 2.5°F hotter WBGT and a 44.7°F hotter ST than the coolest spot (B).
- These spots are located only four blocks (~0.25 miles) away from each other on the same street.
- This underscores the importance of urban tree canopy in mitigating urban heat.

Tree Planting Day

- The heat data was utilised to inform a tree planting initiative hosted by the University of Richmond's Spatial Analysis Lab and the Historic Jackson Ward Association.
- The planting occurred on the morning of November 7th 2025 as a part of the Richmond Tree Week events.
- Canopy shade was expanded along the sidewalks of West Marshall Street and North 2nd Street, some of the hottest identified areas within the Jackson Ward neighborhood.



Figure 11: Image of participating community members during November tree planting



Figure 12 (left): Image of star magnolia planting

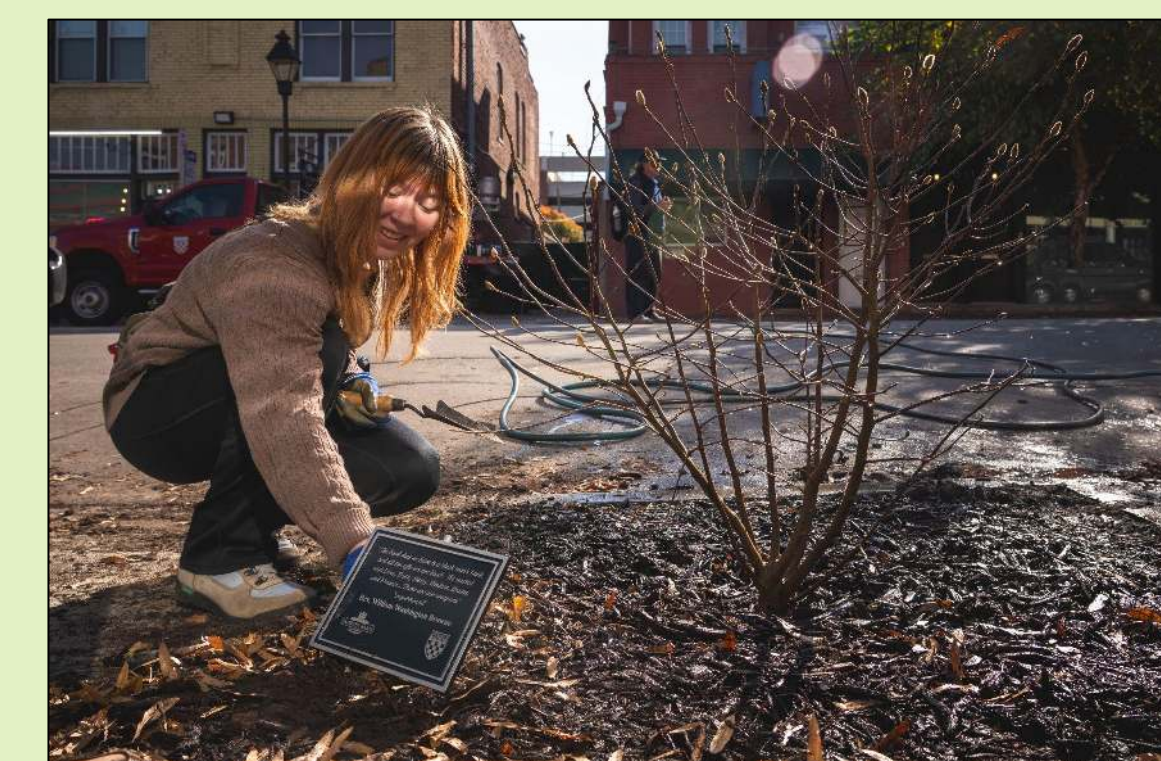


Figure 13 (above): Image of research collaborators from the University of Richmond and the Historic Jackson Ward Association

Figure 14 (left): Image of planted tree with plaque

Acknowledgements

- Janis Allen, President of the Historic Jackson Ward Association, for connecting us with the community and supporting our partnership
- Beth Zizzamia and Kyle Redican for the opportunity to work on this project and their guidance
- The University of Richmond's Department of Geography, Environment, and Sustainability and the School of Arts and Sciences for funding this research project

References

